ENV 790.30 - Time Series Analysis for Energy Data | Spring 2025

Assignment 3 - Due date 02/03/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A03_Sp25.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

Please keep this R code chunk options for the report. It is easier for us to grade when we can see code and output together. And the tidy.opts will make sure that line breaks on your code chunks are automatically added for better visualization.

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

Questions

Consider the same data you used for A2 from the spreadsheet "Table_10.1_Renewable_Energy_Production_and_Consumpti
The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. This time you will work only with the following columns: **Total Renewable Energy Production**; and **Hydroelectric Power Consumption**.

Create a data frame structure with these two time series only.

R packages needed for this assignment:"forecast","tseries", and "Kendall". Install these packages, if you haven't done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
#Load/install required package here
library(forecast)
library(tseries)
library(dplyr)
library(Kendall)

# additional packages
library(readxl)
library(lubridate)
library(ggplot2)
library(glue)
library(tidyverse)
```

Read in the raw data

```
filepath <- "./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx"

# load the data
energy_data_raw <- read_excel(filepath, skip = 11)
```

```
## New names:
## * `` -> `...1`
## * `(Trillion Btu)` -> `(Trillion Btu)...2`
## * `(Trillion Btu)` -> `(Trillion Btu)...3`
## * `(Trillion Btu)` -> `(Trillion Btu)...4`
## * `(Trillion Btu)` -> `(Trillion Btu)...5`
## * `(Trillion Btu)` -> `(Trillion Btu)...6`
## * `(Trillion Btu)` -> `(Trillion Btu)...7`
## * `(Trillion Btu)` -> `(Trillion Btu)...8`
## * `(Trillion Btu)` -> `(Trillion Btu)...9`
## * `(Trillion Btu)` -> `(Trillion Btu)...10`
## * `(Trillion Btu)` -> `(Trillion Btu)...11`
## * `(Trillion Btu)` -> `(Trillion Btu)...12`
## * `(Trillion Btu)` -> `(Trillion Btu)...13`
## * `(Trillion Btu)` -> `(Trillion Btu)...14`
```

```
# read in column names
colnames <- read_excel(filepath, skip = 10) %>% names()
names(energy_data_raw) <- colnames  # assign colnames to raw dataset
```

Basic data processing

```
# clean up names and select columns of interest
energy_data <- energy_data_raw %>%
  janitor::clean_names() %>%
  select(month, total_renewable_energy_production, hydroelectric_power_consumption)
```

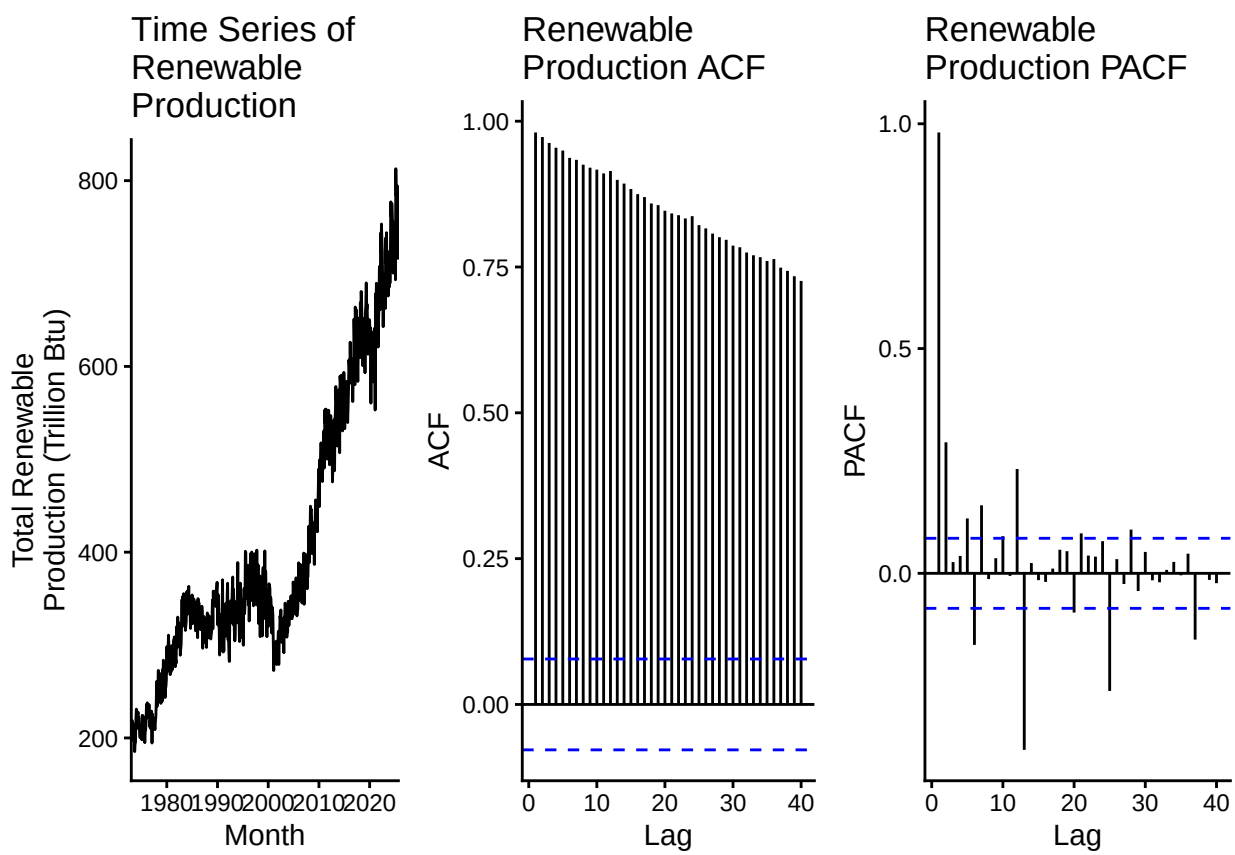
##Trend Component

Q1

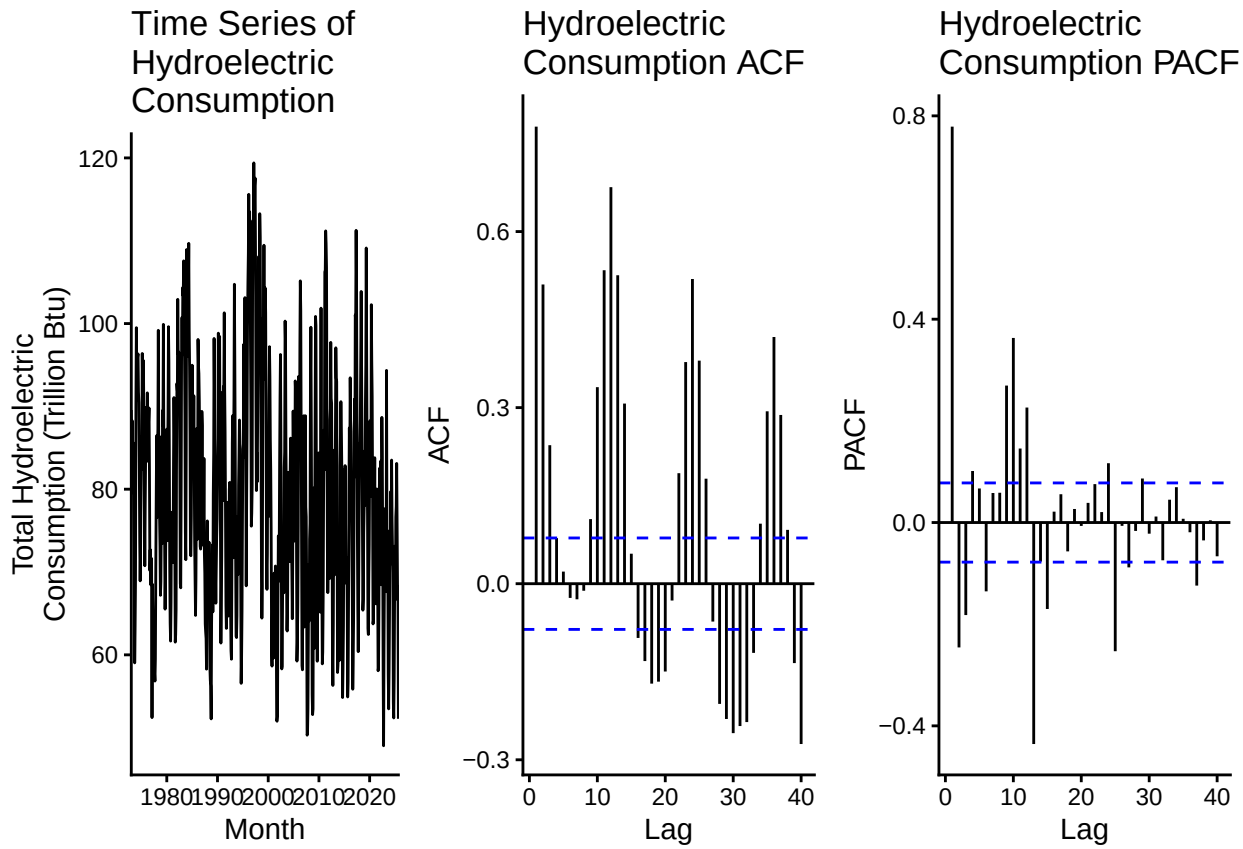
For each series (Total Renewable Production and Hydroelectric Consumption) create three plots arranged in a row (side-by-side): (1) time series plot, (2) ACF, (3) PACF. Use cowplot::plot_grid() to place them in a grid.

```
# function to create all three plots for a given series
create_plots <- function(series, description){
  time_series_plot <- ggplot(energy_data,
                             aes(x = month, y = series)) +
    geom_line() +
    scale_x_date(expand = c(0,0)) +   # compress x axis
    theme_classic() +
    labs(x = "Month",
         y = glue("Total {description} (Trillion Btu)"),
         title = glue("Time Series of\n{description}"))
```

```r
  acf_plot <- ggAcf(series, lag.max = 40) +
    theme_classic() +
    labs(title = glue("{description} ACF"))

  pacf_plot <- ggPacf(series, lag.max = 40) +
    theme_classic() +
    labs(title = glue("{description} PACF"))

  cowplot::plot_grid(time_series_plot, acf_plot, pacf_plot,
                     ncol = 3)
}

# Series 1: Renewable Production
create_plots(energy_data$total_renewable_energy_production,
             "Renewable\nProduction")
```




```r
# Series 2: Hydroelectric Consumption
create_plots(energy_data$hydroelectric_power_consumption,
             "Hydroelectric\nConsumption")
```



Q2

From the plot in Q1, do the series Total Renewable Energy Production and Hydroelectric Power Consumption appear to have a trend? If yes, what kind of trend?

> Answer: Renewable Energy Production appears to have a strong positive linear trend over time, although this trend seems to stall over the period 1985–2000. Hydroelectric Power Consumption does not not appear to have a strong trend. There may be a slight negative trend, but this is difficult to tell apart from the seasonal variation.

Q3

Use the *lm()* function to fit a linear trend to the two time series. Ask R to print the summary of the regression. Interpret the regression output, i.e., slope and intercept. Save the regression coefficients for further analysis.

```
t <- c(1:nrow(energy_data))  # time variable for linear model

# Total Renewable Energy Production
linear_model_renewable <- lm(energy_data$total_renewable_energy_production~t)
summary(linear_model_renewable)
```

```
##
## Call:
```

```
## lm(formula = energy_data$total_renewable_energy_production ~
##     t)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -154.81  -39.55   12.52   41.49  171.15
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 171.44868    5.11085   33.55   <2e-16 ***
## t             0.74999    0.01397   53.69   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.22 on 631 degrees of freedom
## Multiple R-squared:  0.8204, Adjusted R-squared:  0.8201
## F-statistic:  2883 on 1 and 631 DF,  p-value: < 2.2e-16
```

```r
# save coefficients
renewable_b0 <- linear_model_renewable$coefficients[1] %>% as.numeric()
renewable_b1 <- linear_model_renewable$coefficients[2] %>% as.numeric()

# Hydroelectric Power Consumption
linear_model_hydro <- lm(energy_data$hydroelectric_power_consumption~t)
summary(linear_model_hydro)
```

```
##
## Call:
## lm(formula = energy_data$hydroelectric_power_consumption ~ t)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -30.190 -10.214  -0.715   8.909  39.723
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 83.223802   1.110552  74.939  < 2e-16 ***
## t           -0.012199   0.003035  -4.019 6.55e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.95 on 631 degrees of freedom
## Multiple R-squared:  0.02496,    Adjusted R-squared:  0.02342
## F-statistic: 16.15 on 1 and 631 DF,  p-value: 6.547e-05
```

```r
# save coefficients
hydro_b0 <- linear_model_hydro$coefficients[1] %>% as.numeric()
hydro_b1 <- linear_model_hydro$coefficients[2] %>% as.numeric()
```

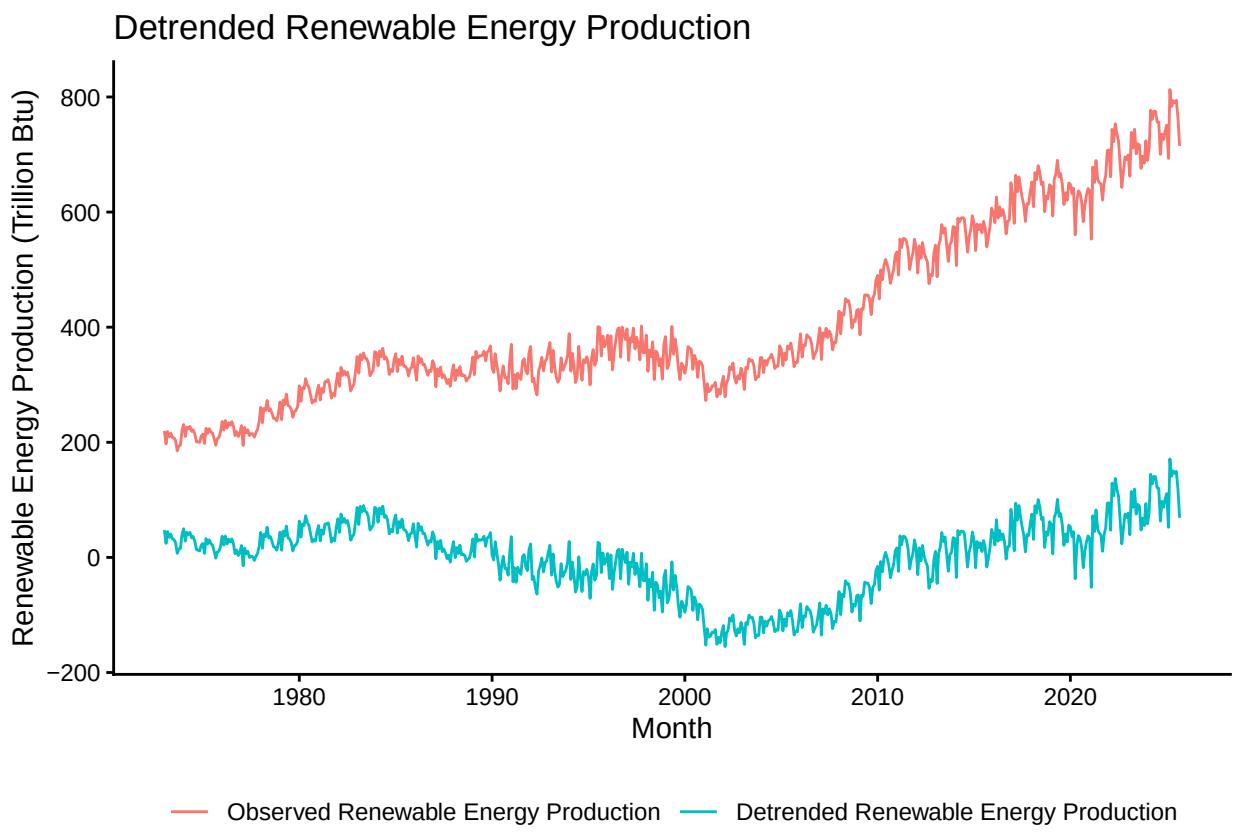
Interpretation: The linear trend fit to the renewable energy production data is positive with an adjusted r-squared value of 0.8201, meaning 82% of variation in this series can be explained by the linear trend over time. These results are significant with a p-value less than 0.05. The linear trend fit to the hydroelectric consumption data is negative with an adjusted r-squared value of

0.02343, meaning only 2.34% of variation in the data can be explained by this trend, but that a trend is present and significant with a p-value < 0.05.

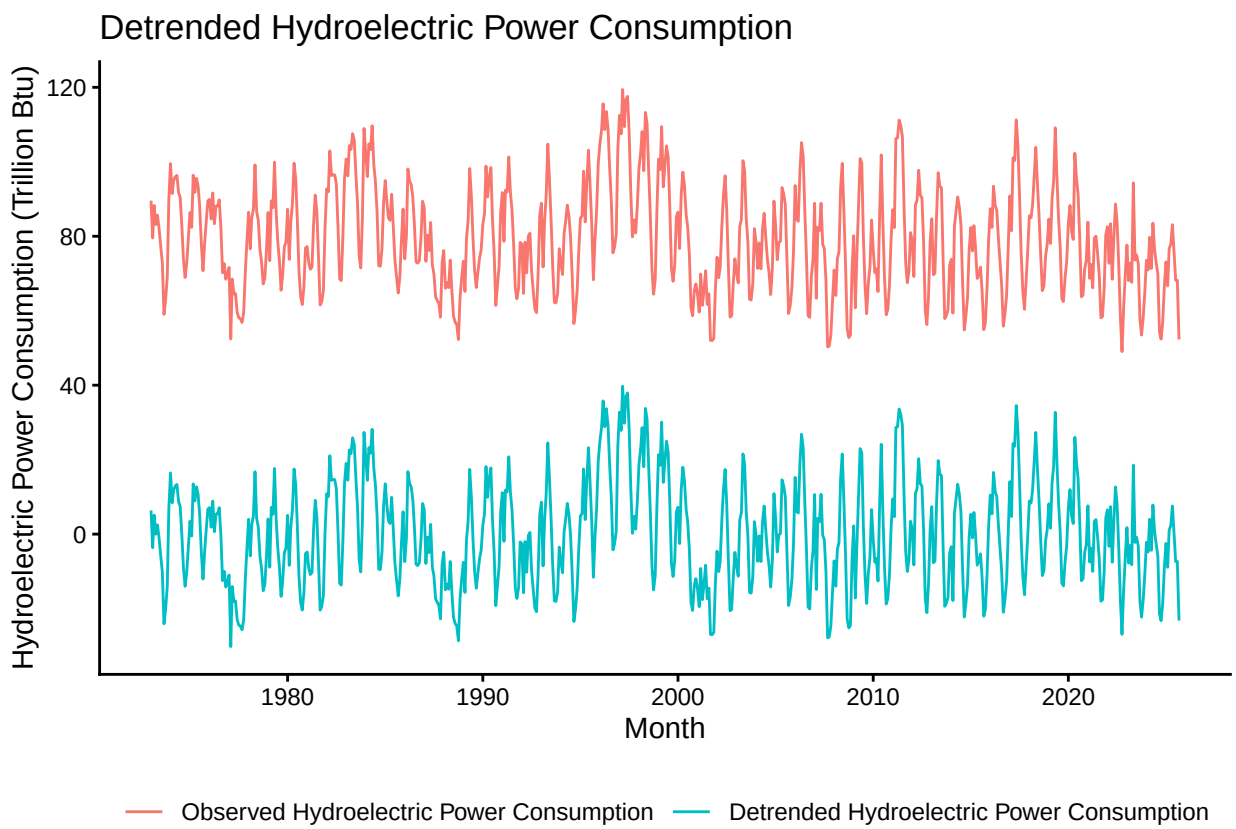
Q4

Use the regression coefficients to detrend each series (subtract fitted linear trend). Plot detrended series and compare with the original time series from Q1. Describe what changed.

```r
# function to detrend series
detrend_series <- function(series, b0, b1, description){
  # subtract linear trend
  series_detrend <- series - (b0 + b1*t)

  # x-axis values
  months <- seq(first(energy_data$month), last(energy_data$month),
                by = "month")

  # combine values into a df to plot
  df_to_plot <- data.frame("month" = months,
                           "series" = series,
                           "series_detrend" = series_detrend) %>%
    pivot_longer(cols = c("series", "series_detrend"),
                 names_to = "series") %>%
    mutate(series = factor(series,
                           levels = c("series", "series_detrend"),
                           labels = c(glue("Observed {description}"),
                                      glue("Detrended {description}"))))
  # plot both time series
  time_series_plot <- ggplot(df_to_plot) +
    geom_line(aes(x = month, y = value, color = series)) +
    theme_classic() +
    theme(legend.position = "bottom") +
    labs(x = "Month",
         y = glue("{description} (Trillion Btu)"),
         title = glue("Detrended {description}"),
         color = NULL)

  return(list(series_detrend, time_series_plot))
}

# detrend renewable energy production
renewable_detrended <- detrend_series(energy_data$total_renewable_energy_production,
                                      b0 = renewable_b0, b1 = renewable_b1,
                                      description = "Renewable Energy Production")

renewable_detrend_series <- renewable_detrended[[1]] # save detrended series object
renewable_detrended[[2]]   # show plot
```




```r
# detrend hydroelectric power consumption
hydro_detrended <- detrend_series(energy_data$hydroelectric_power_consumption,
                                  b0 = hydro_b0, b1 = hydro_b1,
                                  description = "Hydroelectric Power Consumption")

hydro_detrend_series <- hydro_detrended[[1]]   # save detrended series object
hydro_detrended[[2]]   # show plot
```



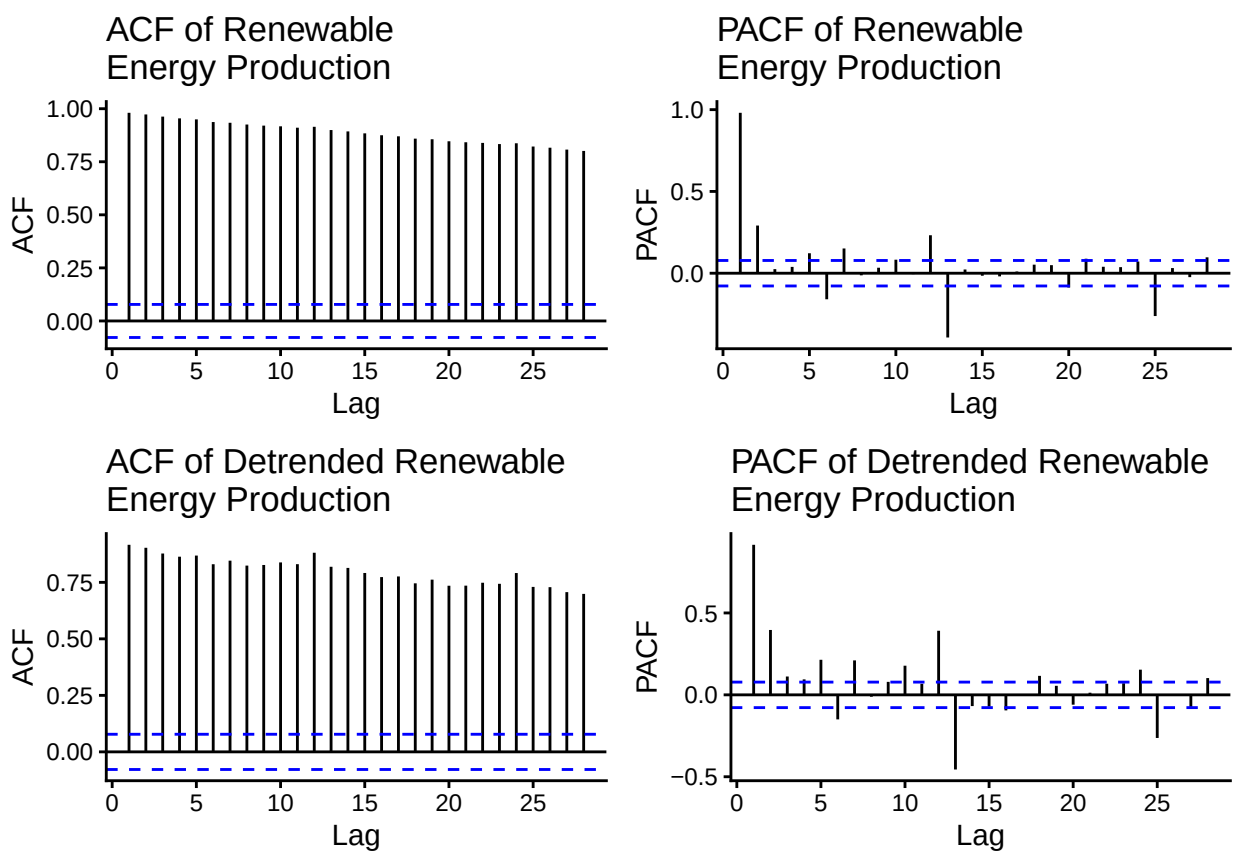
Answer: Both detrended series no longer have a trend component and their values are centered around 0. For hydroelectric power consumption, the shape of the detrended series appears the same and only the values have been shifted down. The series of renewable energy production appears to have been flattened out.

Q5

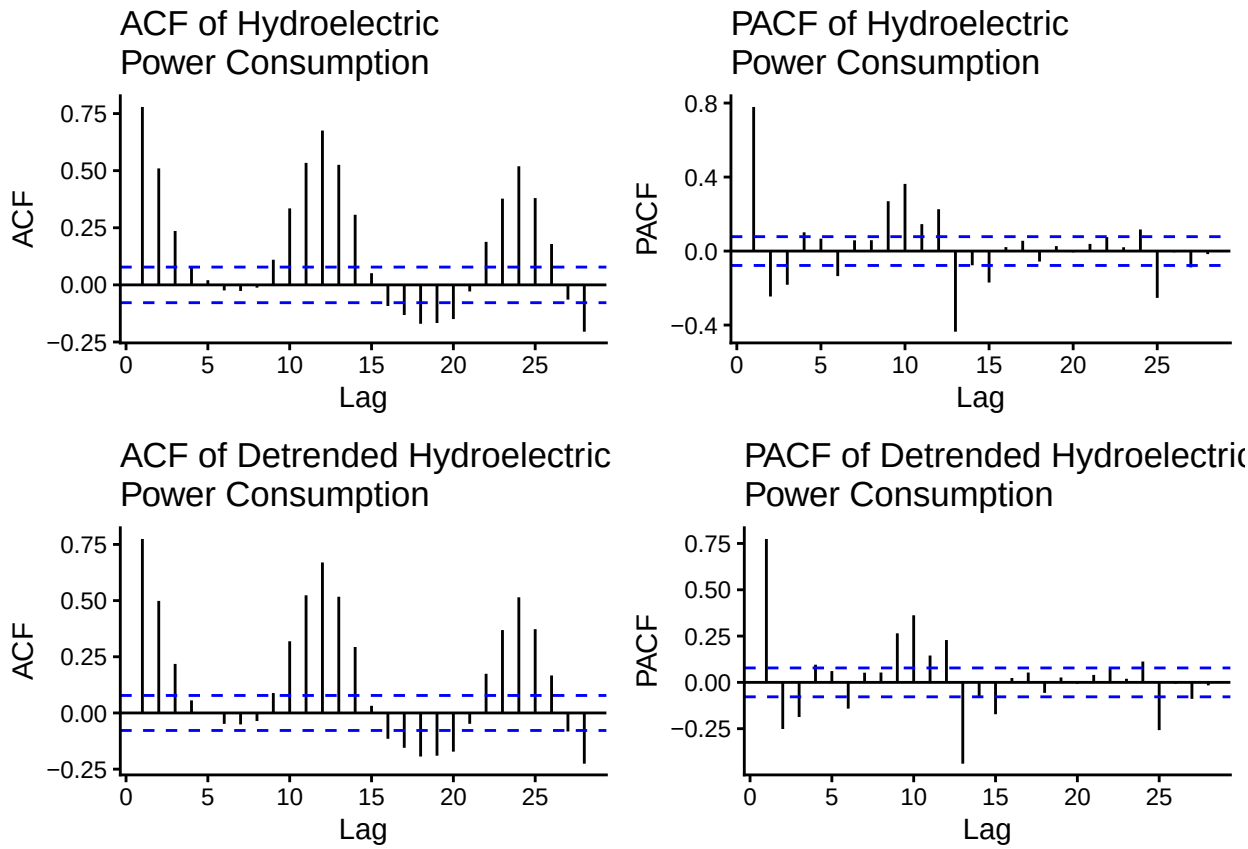
Plot ACF and PACF for the detrended series and compare with the plots from Q1. You may use plot_grid() again to get them side by side to make it easier to compare. Did the plots change? How?

```
plot_detrend_acf <- function(series, detrend_series, description, trend){
  plot_acf <- ggAcf(series) +
    theme_classic() +
    labs(title = glue("ACF of {description}"))

  plot_pacf <- ggPacf(series) +
    theme_classic() +
    labs(title = glue("PACF of {description}"))

  plot_acf_detrend <- ggAcf(detrend_series) +
    theme_classic() +
    labs(title = glue("ACF of {trend} {description}"))

  plot_pacf_detrend <- ggPacf(detrend_series) +
    theme_classic() +
```

```r
    labs(title = glue("PACF of {trend} {description}"))

  cowplot::plot_grid(plot_acf, plot_pacf, plot_acf_detrend, plot_pacf_detrend,
            ncol = 2, nrow = 2)
}
# Renewable Energy Production
plot_detrend_acf(energy_data$total_renewable_energy_production,
                 renewable_detrend_series,
                 "Renewable\nEnergy Production", "Detrended")
```




```r
# Hydroelectric Power Consumption
plot_detrend_acf(energy_data$hydroelectric_power_consumption,
                 hydro_detrend_series,
                 "Hydroelectric\nPower Consumption", "Detrended")
```



Answer: There is a slight change in the ACF and PACF for the renewable energy production detrended series. For these data, the seasonal component is a bit more pronounced in the ACF and PACF of the detrended series where there appears to be a slight spike in autocorrelation at around 12 lags. There appears to be no change in the ACF and PACF for the detrended hydroelectric power consumption series.

Seasonal Component

Set aside the detrended series and consider the original series again from Q1 to answer Q6 to Q8.

Q6

Just by looking at the time series and the acf plots, do the series seem to have a seasonal trend? No need to run any code to answer your question. Just type in you answer below.

Answer: For renewable energy production, there is no apparent seasonal component to the time series and ACF plots. For hydroelectric power consumption, there is a very clear seasonal component to both the ACF and time series plots.

Q7

Use function *lm()* to fit a seasonal means model (i.e. using the seasonal dummies) to the two time series. Ask R to print the summary of the regression. Interpret the regression output. From the results, which series have a seasonal trend? Do the results match you answer to Q6?

```r
# create time series objects
start <- c(first(year(energy_data$month)), first(month(energy_data$month)))

renewable_ts <- ts(energy_data$total_renewable_energy_production,
                   start = start, frequency = 12)
hydro_ts <- ts(energy_data$hydroelectric_power_consumption,
               start = start, frequency = 12)

# create seasonal dummies
dummies <- seasonaldummy(renewable_ts)

# fit seasonal means model to both series
# Renewable Energy Production
renewable_seasonal_model <- lm(renewable_ts ~ dummies)
summary(renewable_seasonal_model)
```

```
##
## Call:
## lm(formula = renewable_ts ~ dummies)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -213.33  -97.36  -59.88  121.55  389.62
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  417.265     21.096  19.779   <2e-16 ***
## dummiesJan     2.090     29.693   0.070    0.944
## dummiesFeb   -34.524     29.693  -1.163    0.245
## dummiesMar     5.956     29.693   0.201    0.841
## dummiesApr    -6.900     29.693  -0.232    0.816
## dummiesMay     8.162     29.693   0.275    0.784
## dummiesJun    -2.231     29.693  -0.075    0.940
## dummiesJul     3.864     29.693   0.130    0.897
## dummiesAug    -3.978     29.693  -0.134    0.893
## dummiesSep   -29.033     29.693  -0.978    0.329
## dummiesOct   -19.937     29.834  -0.668    0.504
## dummiesNov   -20.617     29.834  -0.691    0.490
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 152.1 on 621 degrees of freedom
## Multiple R-squared:  0.008243,   Adjusted R-squared:  -0.009324
## F-statistic: 0.4692 on 11 and 621 DF,  p-value: 0.9223
```

```r
# Hydro Power Consumptions
hydro_seasonal_model <- lm(hydro_ts ~ dummies)
summary(hydro_seasonal_model)
```

```
##
## Call:
## lm(formula = hydro_ts ~ dummies)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -30.895  -6.368  -0.595   6.213  32.557
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   79.724      1.436  55.511  < 2e-16 ***
## dummiesJan     4.951      2.021   2.449 0.014591 *
## dummiesFeb    -2.415      2.021  -1.195 0.232608
## dummiesMar     7.116      2.021   3.520 0.000463 ***
## dummiesApr     5.614      2.021   2.777 0.005649 **
## dummiesMay    14.080      2.021   6.965 8.38e-12 ***
## dummiesJun    10.780      2.021   5.333 1.36e-07 ***
## dummiesJul     4.003      2.021   1.980 0.048091 *
## dummiesAug    -5.320      2.021  -2.632 0.008710 **
## dummiesSep   -16.598      2.021  -8.211 1.28e-15 ***
## dummiesOct   -16.329      2.031  -8.040 4.56e-15 ***
## dummiesNov   -10.782      2.031  -5.308 1.54e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.36 on 621 degrees of freedom
## Multiple R-squared:  0.4714, Adjusted R-squared:  0.4621
## F-statistic: 50.35 on 11 and 621 DF,  p-value: < 2.2e-16
```

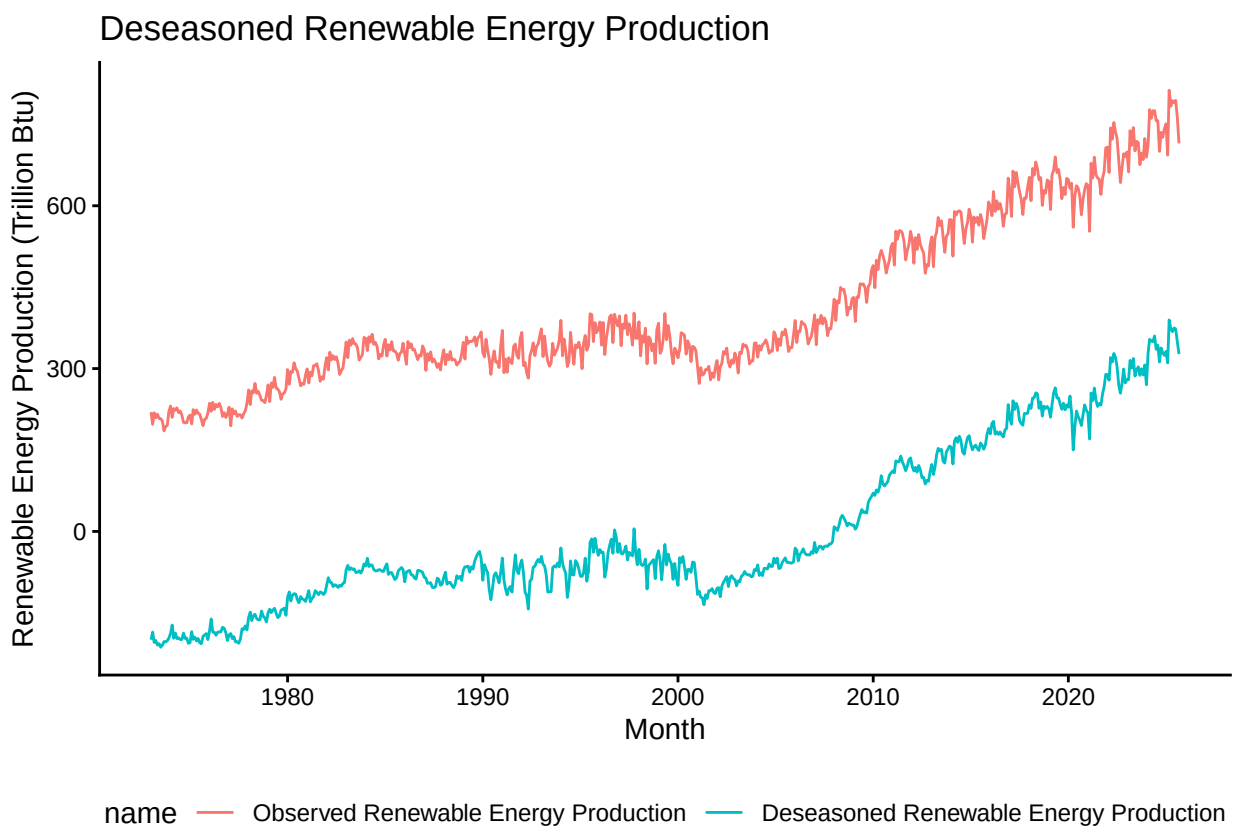
Answer: The seasonal model of the renewable energy production series has an adjusted r-squared value of -0.009 and a p-value of 0.9223. My understanding of a negative value for adjust r-squared is that the seasonal dummies are an irrelevant predictor and that the model fits the data worse than a simple horizontal line at the mean. The seasonal model of the hydroelectric power consumption has an adjusted r-squared value of 0.4621 and a p-value < 0.05, meaning the model is significant and can account for 46.2% of variation in the data. This is in-line with my assessment above that there is no seasonal component to renewable production, but there is a strong seasonal component to hydroelectric consumption.

Q8

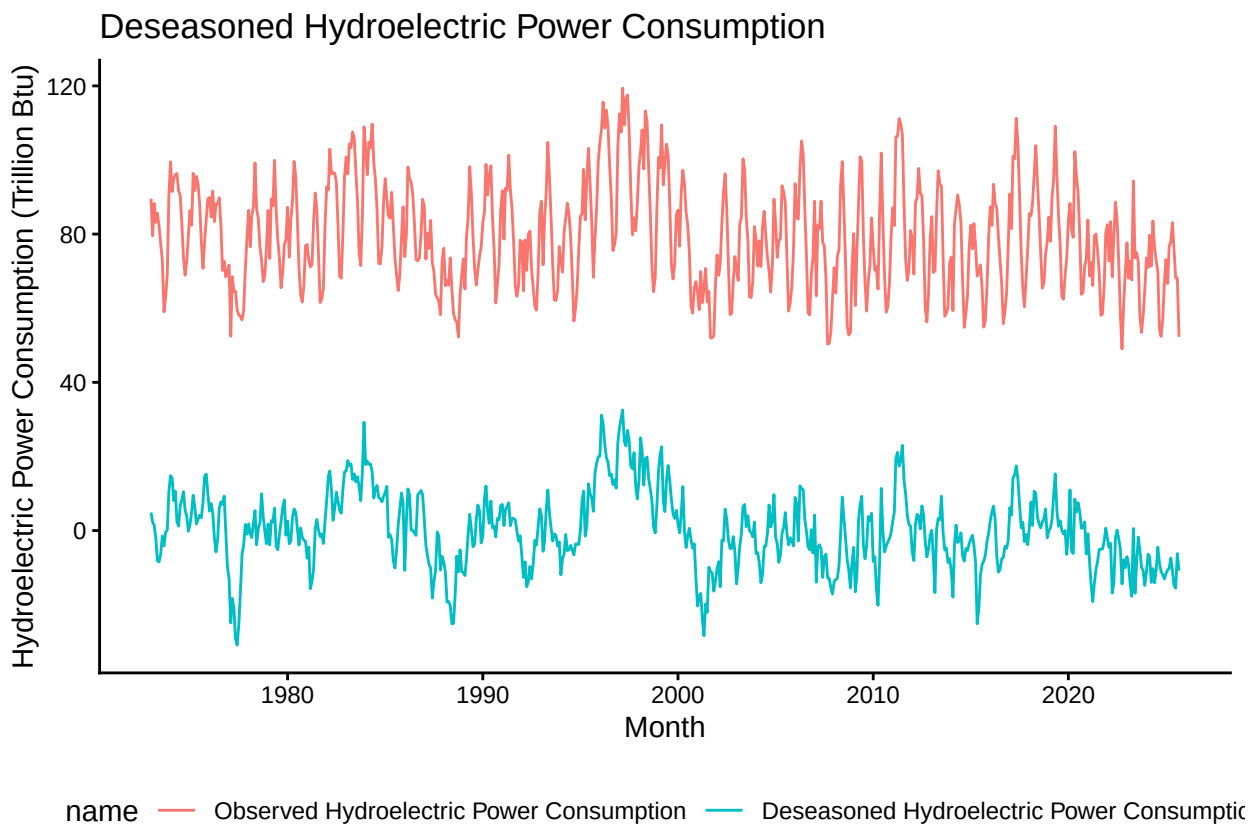
Use the regression coefficients from Q7 to deseason the series. Plot the deseason series and compare with the plots from part Q1. Did anything change?

```r
deseason_series <- function(series, seasonal_model, description=""){
  beta_int <- seasonal_model$coefficients[1]   # intercept term
  beta_coeff <- seasonal_model$coefficients[2:12]   # all other coefficients
  #seasonal_component <- beta_int + (coefficients*dummies)

  seasonal_component <- array(0, length(series))  # array of zeros
  for (i in 1:length(series)){
    seasonal_component[i] <- beta_int + (beta_coeff%*%dummies[i,])  # linear product
  }

  # subtract seasonal component from series
  deseasoned_series <- series - seasonal_component

  # data frame for plotting
```

```r
  df_to_plot <- tibble("month" = seq(first(energy_data$month), last(energy_data$month), by = "month"),
                       "series" = series,
                       "deseasoned_series" = deseasoned_series) %>%
    pivot_longer(cols = c("series":"deseasoned_series")) %>%
    mutate(name = factor(name,
                         levels = c("series", "deseasoned_series"),
                         labels = c(glue("Observed {description}"), glue("Deseasoned {description}"))))

  # plot deseasoned series with normal series
  plot <- ggplot(df_to_plot) +
    geom_line(aes(x = month, y = value, color = name)) +
    theme_classic() +
    theme(legend.position = "bottom") +
    labs(x = "Month",
         y = glue("{description} (Trillion Btu)"),
         title = glue("Deseasoned {description}"))

  return(list(deseasoned_series, plot))
}

renewable_deseasoned <- deseason_series(energy_data$total_renewable_energy_production,
                                        renewable_seasonal_model, "Renewable Energy Production")
renewable_deseasoned[[2]]
```



```
hydro_deseasoned <- deseason_series(energy_data$hydroelectric_power_consumption,
                                    hydro_seasonal_model, "Hydroelectric Power Consumption")
hydro_deseasoned[[2]]
```

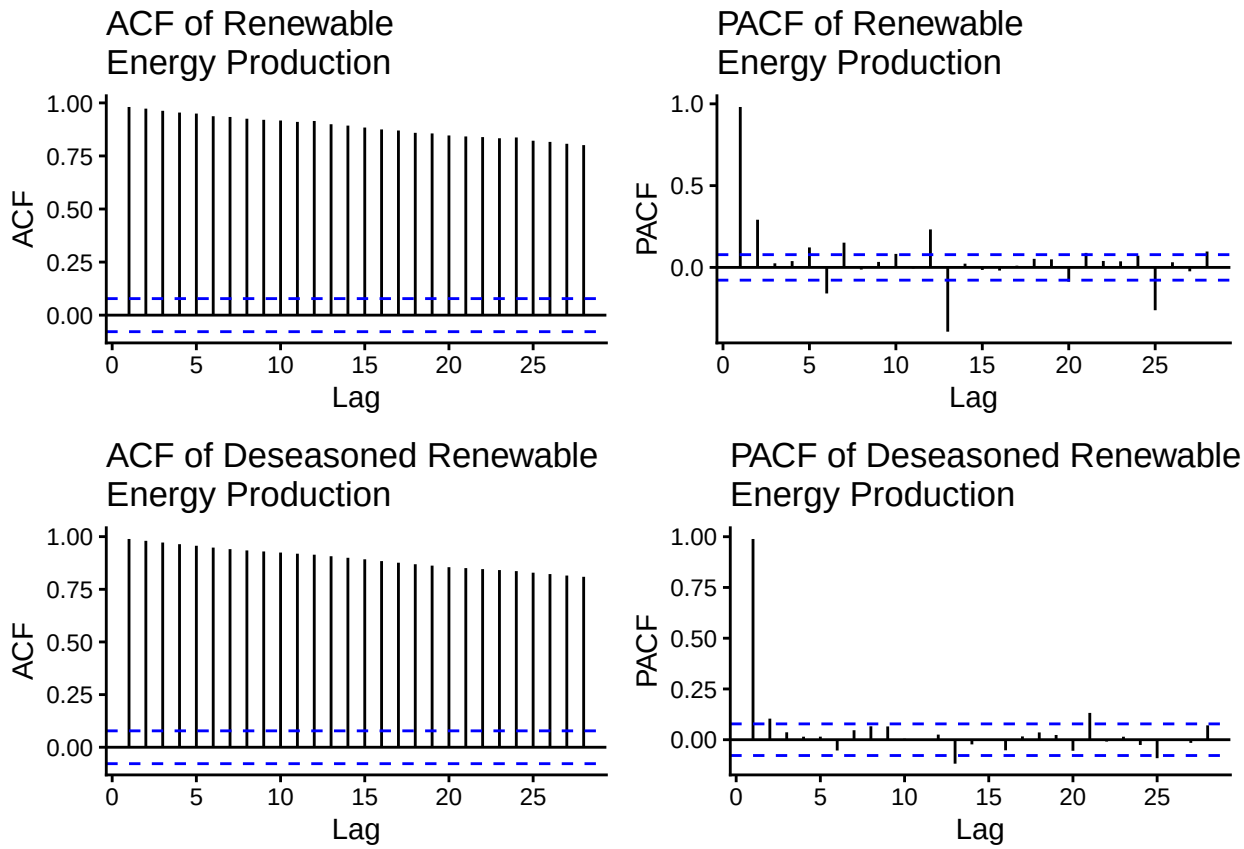


Answer: The above plots compare the time series plot of the observed data to the series after removing the seasonal component. For both de-seasoned series, the mean is now zero. For the renewable energy series, there was not much change to the series except for a slight flattening of some of the fluctuation in the observed data. For the hydroelectric power series, the seasonal up-and-down movement has been removed, and now we only see the overal trending increases/decreases in the observed data.

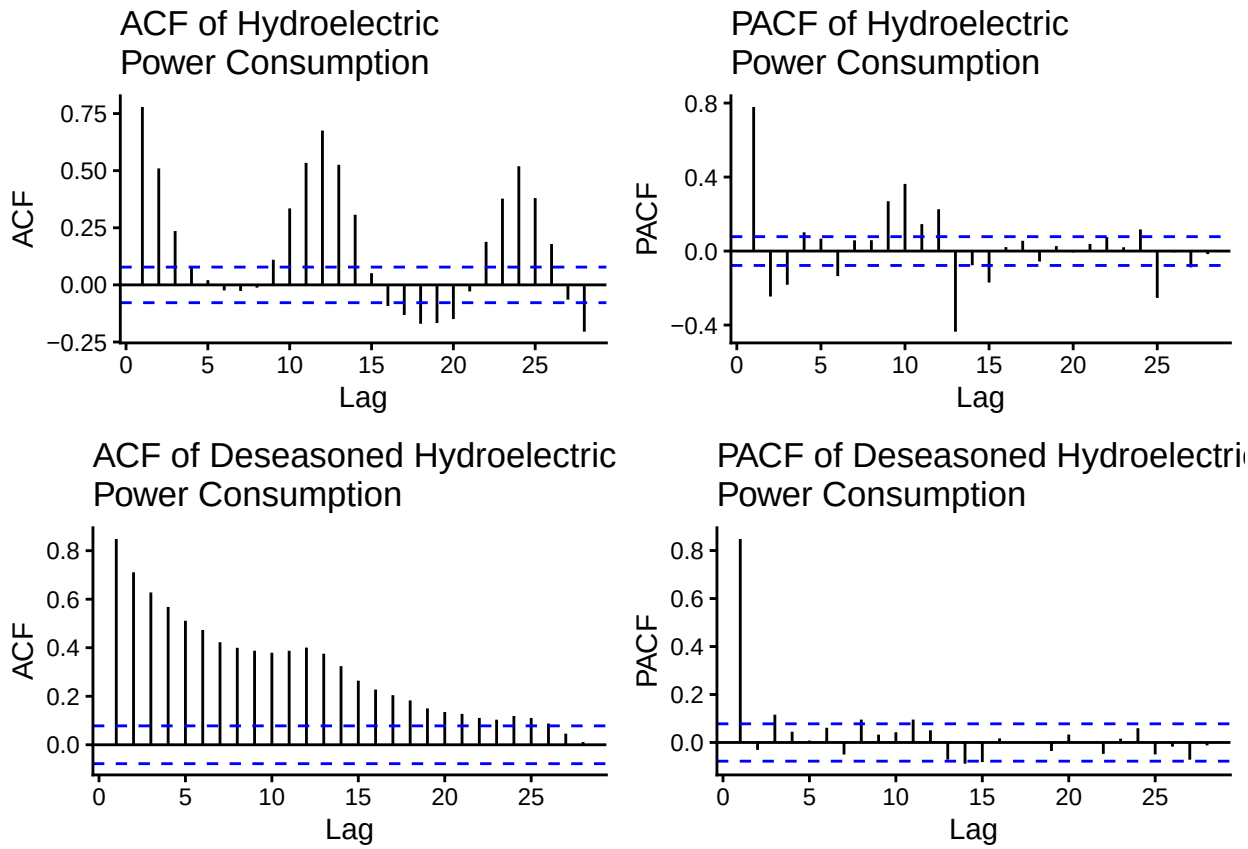
Q9

Plot ACF and PACF for the deseason series and compare with the plots from Q1. You may use plot_grid() again to get them side by side. Did the plots change? How?

```
# using function defined above to plot ACF / PACF of deseasoned and observed series
plot_detrend_acf(energy_data$total_renewable_energy_production,
                 renewable_deseasoned[[1]],
                 "Renewable\nEnergy Production", "Deseasoned")
```




```
plot_detrend_acf(energy_data$hydroelectric_power_consumption,
                 hydro_deseasoned[[1]],
                 "Hydroelectric\nPower Consumption", "Deseasoned")
```



Answer: There is very little change to the ACF of the observed and deseasoned renewable energy production series. You can see some difference in the PACF plot where the PACF is insignificant for almost all lags after the first lag and there is no longer a spike in PACF around 12 lags. There is a greater impact on the ACF/PACF of the hydroelectric series. You no longer see the sinosoidal pattern in the ACF plot and the ACF values for the deseasoned series decay much more rapidly. Almost all PACF values for the deseasoned series are insignificant after the first lag.